

NOORELDEEN ABDELKARIM

Applied AI and Machine Learning Engineering

Cairo, Egypt | Seeking fully remote roles from Egypt

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Education

German University in Cairo

2022 - Expected Jun 2027

B.Sc. in Media Engineering and Technology | Computer Science Engineering major

Profile

Computer Science Engineering student with PyTorch neural-network research and contributions to Sahahly's AI grading and learning platforms. Experience spans document grading, Python APIs, human review, background processing and full-stack delivery.

Experience

Sahahly

Mar 2026 - Present

Applied AI and Full-Stack Development Contributor

- Adapted and integrated question-paper/mark-scheme indexing into a shared multimodal grading platform using React, Node.js, Express and MongoDB; validated structured grading outputs.
- Added recovery for malformed or oversized model responses, missing source pages and resumable batch finalization to preserve grading results through failures.
- Extended the LMS through Django REST APIs and Next.js interfaces for bulk Gemini grading, configurable model selection, student resubmissions and version history.
- Added database/cache health checks and monitoring configuration; contributed regression coverage for reopening reviewed grades and preserving handwritten PDF annotations.

Orascom Construction

Jul 2025

IT and Software Engineering Intern

- Independently developed a MERN construction-site management prototype from manager requirements, modeling sites, towers, floors and tasks.
- Trained a scikit-learn Random Forest classifier on 100+ synthetic task records and served material-waste risk predictions through a FastAPI microservice.

Research and Projects

PINNs Thesis and Heat-Equation Extension | GUC | A+ | [Code](#)

2026

- Implemented PyTorch PINNs and matched neural-network baselines across 10 controlled studies of 8 synthetic ODE systems, using RK4 reference trajectories and automatic differentiation.
- Investigated physics-loss weighting, architecture, training budgets, sparse data and noise; evaluated MSE and seed sensitivity to identify where physical constraints helped or degraded predictions.
- Extended the study to a synthetic 1D heat-equation benchmark: 15 paired runs across five seeds yielded 89.6% lower mean full-grid MSE than a data-only neural network.

Technical Skills

AI and ML: Python, PyTorch, scikit-learn, NumPy, pandas, PINNs, model evaluation, LLM APIs, structured outputs

Software: JavaScript, React, Next.js, Node.js, Express, Django REST Framework, FastAPI, Git, REST APIs

Data and reliability: SQL, PostgreSQL, MongoDB, Celery, Redis, regression testing, Docker Compose monitoring

Certifications and Training

IBM Data Science Professional Certificate | Completed 5 Oct 2026 | 12 courses

Additional Experience

Schneider Electric | Performance and Data Analytics Internship, Jul - Aug 2026. Sika Egypt | IT and Data Science Internship, Aug 2025.

Arabic: native | English: fluent | German: basic